Automated Medical Image Analysis for Brain Tumor and Skin Disease Diagnosis Using Transfer Learning and Grad-CAM

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***Abstract*—Timely and accurate categorization of medical images is essential for improving patient care and aiding in the decision-making processes in hospitals. We suggest a deep learning system that detects brain and skin pathologies using MRI and dermoscopy elements. The brain tumor model was tested on T1-weighted contrast-enhanced MRIs. We created a dataset of 7,200 MRI images, including gliomas, meningiomas, pituitary tumors, and a control group. We experimented with our custom-made CNN and transfer learning methods: VGG16 and ResNet-50. The analysis of the results showed that VGG16 produces superior results in terms of accuracy of testing (94% accuracy) in comparison to ResNet-50 (93%) and the custom CNN (91%). The analysis of skin diseases was conducted based on 9,548 dermoscopy images containing five classifications of the diseases: acne, vitiligo, hyperpigmentation, nail psoriasis, and Steven-Johnson Syndrome/Toxic Epidermal Necrolysis. In this case, the architectures EfficientNetB0, EfficientNetB3, and MobileNetV2 were worked with, with the highest result recorded for EfficientNetB0 of 99% accuracy and AUC equal to 0.993. To improve the transparency and clinical applicability of the framework, we used Gradient-weighted Class Activation Mapping (Grad-CAM) to gain insight into important parts of the input images that contributed to the model predictions. The results prove the value of the combination of CNNs and transfer learning across many medical image processing applications ensuring efficiency and generalization. Finally, the proposed highly interpretable method is an efficient and scalable support tool for doctors.**

***Index Terms*—Brain Tumor Classification, Skin Disease Classification, Deep Learning, Transfer Learning, Explainable AI (XAI), Computer-Aided Diagnosis (CAD), Convolutional Neural Networks (CNN), Grad-CAM**

I. Introduction

Every year, millions of lives are at stake as doctors dash against the clock to find out what is wrong when they can't see what is wrong. Two of the most threatening of these are brain tumors and skin diseases, two diseases with vast differences in nature, but with a common and serious characteristic – their difficulty in early detection and the classification of them without specialized knowledge. Primary brain tumors make up around 308,000 new cases a year worldwide, and gliomas account for almost 80% of all malignant brain tumor diagnoses, according to the World Health Organization (WHO). In the meantime, skin diseases are estimated to affect 1.9 billion people around the world with some, like Stevens-Johnson Syndrome (SJS) and Toxic Epidermal Necrolysis (TEN), having death rates up to 30%. The difference between life and death in both domains is a single point – quick and accurate diagnosis of the disease.

Both conditions follow a traditional diagnostic process that heavily relies on highly trained specialists for interpretation of the MRI scans (neuro-radiologists) and dermoscopic images (dermatologists). However, in low and middle income countries, the specialist to patient ratio is severely skewed, with many people unable to get a timely accurate diagnosis [2]. Human error isn't a rarity, but a reality when a single neuroradiologist is interpreting hundreds of MRI scans every day. Likewise, there is an acute shortage of dermatologists throughout the world, and skin lesions may be overlooked for months, resulting in treatable lesions becoming life-threatening emergencies.

Deep learning is now stepping in with promising results [3] into this space. Convolutional Neural Networks (CNNs) have dramatically changed the ways that medical images have been analyzed in the past, by learning hierarchical representations directly from raw pixel data and identifying subtle morphological patterns that are not detectable by the human eye. This capability has been further enhanced by transfer learning, which involves training a model on a large data set, such as ImageNet, and then fine-tuning it for a specific application, such as medical imaging, thus saving a lot of labeled data and improving classification accuracy. Some models, like VGG16, ResNet-50 or EfficientNet have shown to outperform clinical experts in a number of disease domains [1] [4].

However, there are still some major issues to address. Brain MRI and dermoscopic images, mainly due to the high intra-class variability and high inter-class similarity, make robust classification an inherent difficult task. Moreover, the interpretability of the model is still a major hurdle for clinical use [5]: a black-box model that makes accurate predictions

but provides no explanation is not very useful for a physician who needs to explain a diagnosis. To meet these challenges, high-performing architectures are necessary, alongside careful preprocessing strategies and explanations mechanisms that provide visual information that is clinically relevant based on the model's decisions.

To remedy this, the present paper introduces a deep learning based architecture for the classification of brain tumor MRI and multi-class skin disease detection. They are tested on 7,200 MRI scans across four tumor types using three architectures (custom CNN, VGG16, ResNet-50) and 9,548 dermoscopic images from five skin disease classes using EfficientNetB0, EfficientNetB3 and MobileNetV2. Grad-CAM visualizations are embedded in the system to assure that predictions not only are accurate but also are visually accountable at a pixel-by-pixel level.

The key contributions of this paper are summarized as follows:

- A single coherent pipeline is created to solve both the problem of brain tumour classification in MRI and multi-class skin disease detection, in a unified deep learning framework.
- The three architectures of custom CNN, VGG16 and ResNet-50 are systematically evaluated for brain tumour classification, where VGG16 with a two stage transfer learning strategy achieves the highest accuracy of 94% results.
- EfficientNetB0, EfficientNetB3, and MobileNetV2 are tested on the five-class skin disease classification task and are found to have an accuracy of 99% and an AUC of 0.993 for EfficientNetB0.
- Both modules are equipped with Grad-CAM to produce visual saliency maps highlighting salient areas with disease relevance, enhancing the model transparency and clinical trustworthiness.
- Specific image preprocessing and augmentation pipelines are developed for each imaging domain; brain MRI is done using TensorFlow/Keras, while skin disease classification is carried out using PyTorch.

II. Related Works

Naqvi et al. [6] presented a survey of deep learning methods for detecting skin cancer, detailing popular CNN architectures, techniques for segmentation and classification, and common benchmarks. The study identifies that the deep learning approaches offer higher diagnosis of skin cancer and improved detection performance, and also address some inherent limitations and research gaps that remain in the field.

Nair et al. [7] proposed a skin cancer detection based on the partitioning of lesions with an improved model based on DeepLabv3+, which is followed by multi-scale Convolutional neural network (MSCNN) models for classification, trained by skin dataset, we trained models with partitioning of lesions (non-segmented/segmented) using skin ISIC 2017 dataset in both segmentation and without Segmentation models and then performed an accuracy comparison, both on different class categories and on the global data classification (Melanoma, Nevus, Seborrheic Keratosis), in which Segmentation gained 4% total classification rate.

Gouda et al. [8] proposed a deep learning method on skin lesions and trained it on the public datasetISIC 2018 by applyESRGAN(Enhanced super-resolution Generative Adversarial Networks)on data augmentation by it and using special CNN model,ResNet50,InceptionV3 and Inception-ResNet models as a transfer learning, then obtained from these models the desired best performance model (which is InceptionV3), it ranked the best and obtained a rate of accuracy of (85.8).

Pacheco and Krohling et al. [9] proposed a skin cancer detection methodology that leverages both clinical images and patient clinical features for improved diagnostic effectiveness. This methodology yielded an increased balanced accuracy of nearly 7% in comparison to image-based models by leveraging a recently established dataset and utilizing deep learning models integrating a feature aggregation component.

Chowdary [10] investigated various deep learning models for the detection of diabetic foot ulcers (DFU) utilizing thermal images. Four different architectures, trained and assessed on a publicly accessible thermography database, were evaluated on their performance for identification of DFU, including factors of accuracy, precision, recall, and F1-score, with an ultimate aim to build a stronger and improved model for early DFU detection and computer-aideddiagnosis to assist reduce amputation.

Sharma and Sampath et al. [11] proposed a data augmentation pipeline utilizing MONAI in the BraTS 2020 dataset to improve brain tumor segmentation. These techniques will serve to make an enhancement of robust data augmentation to support U-Net based tumor segmentationmodels' abilities to generalize effectively, particularly considering lack of annotated data and great variance ofglioma dimensions, position and looks.

Wu et al. [12] presented a systematic review of deep learning based skin cancer classification. The review summarized the commonly used CNN architectures, different kinds of dermatological images and available public datasets for skin cancer detection. The review also points out the main difficulties which include imbalanced data, small dataset, domain adaptation and robustness of models. It concludes that future skin cancer classification system should take advantages of lightweight, multimodality and generalizable deep learning models.

Jain and Santhanalakshmi et al. [13] proposed an ensemble deep learning and ensemble learning framework for early brain tumor classification and survival prediction utilizing MRI images. CNN, VGG16, ResNet50 and InceptionV3 models were tested using 7,022 images of MRI for the classification of glioma, meningioma, pituitary and no-tumor. The fine-tuned ResNet50 gained the optimal performance for all models with the overall accuracy of 98.63% which proves its superiority in brain tumor classification and computer-aided diagnosis.

As evidenced from the surveyed literature, deep learning technology has greatly enhance the analysis of medical images

related to the diagnosis of skin cancer, foot diabetic ulcer, and brain tumor. Using CNN-based and transfer learning deep learning models, image segmentation, data augmentation, image enhancement and multimodal feature fusion techniques, state-of-the-art accuracy rates can be achieved to classify diseases in those different medical applications. However, major problems such as scarce annotated dataset, class imbalance, limited generalization ability and complexity of computation persist and are urgently needed to be resolved for achieving high accuracy, generalized, reliable, robust and low-complexity deep learning-based automated disease classification methods.

III. Methodology

Creating an intelligent system from a raw image to a clinically relevant prediction to diagnose disease is not an easy challenge — it's a journey that starts with a raw image and extends all the way to prediction. This proposed architecture is a structured and modular architecture that is capable of dealing with two separate imaging domains: brain tumor classification from MRI scans, and skin disease detection from dermoscopic images. The pipeline is essentially divided into four phases: data ingestion, pre-processing the data for each disease, running inference on the deep learning model and generating explanations using Grad-CAM. The image is processed according to the selected disease module using its specific model pathway and the specific preprocessing stage, resulting in an optimized response for each domain. The brain MRI module is implemented using TensorFlow/Keras, while the skin disease module is implemented using PyTorch, both of which are best suited to the jobs at hand. The result of the system is not only a class label but also a prediction with a visual heatmap showing exactly which parts of the image led the model to decide the class label — a combination that is both accurate and interpretable for clinical use.

A. Dataset Description

All smart systems are only as smart as the information that they are fed. When developing this framework, two well-chosen medical imaging datasets were chosen, one that represents the complexity of the structural complexity of the human brain, and the other the visual variety of skin disease.

1) Brain MRI Dataset: The Brain Tumor MRI dataset provided by Kaggle and curated by Masoud Nickparvar [14] contains 7200 T1-weighted MRI brain scans that are contrast enhanced. The dataset is divided into four clinically different categories: Glioma, Meningioma, Pituitary Tumor and No Tumor. Gliomas are malignant (cancerous) tumors of the glial cells or their precursors, and can be difficult to classify because they do not have a clearly defined boundary and appear very differently from one to another. The meninges that cover the brain give rise to meningiomas; the MRI can be difficult to distinguish between meningiomas and gliomas, and can also be challenging to distinguish from one class to the next. Tumors in the pituitary gland, which sits at the base of the brain, are mostly harmless but must be diagnosed in time to avoid complications from their effects on the hormones and nervous system. The No Tumor class is the negative control, which represents healthy brain scans. The images used in this dataset have varying resolution, orientation and background adding realistic variation to the images, and forcing the trained models to learn more general features and not just artifacts from the particular dataset. This intentional diversity ensures the dataset's suitability for assessing deep learning models designed for real-world clinical applications.

2) Skin Disease Dataset: The Skin Disease dataset was created from various hospital databases and clinical repositories online and is freely accessible in the Mendeley Data Repository [15]. It includes 9548 dermoscopic images in five categories of dermatological diseases. The dataset comprises 1,148 images of acne, one of the most common skin disorders worldwide. Vitiligo is a chronic autoimmune disease that is characterized by depigmented skin patches, and is depicted by 2016 images. The smallest class is hyperpigmentation - an image with an excess of melanin - that adds 700 images to the dataset, creating a meaningful class imbalance challenge. Images of Nail Psoriasis, a chronic inflammatory disease of the nails and skin make up 2,520 images. The largest class of images, 3,164, is that of a rare but life-threatening mucocutaneous reaction, Stevens-Johnson Syndrome / Toxic Epidermal Necrolysis (SJS-TEN). The addition of SJS-TEN to this list is especially noteworthy as it has a high mortality – and urgent need for timely diagnosis. There is natural variation in the image quality, lighting conditions and presentation of the lesions, which makes the data set a realistic and difficult benchmark for models of skin disease classification.

B. Image Preprocessing

1) Brain Tumor MRI Image Preprocessing: The MRI brain images were resized to fit the architecture they were initially intended for. To reduce the computational complexity during the training stage, all images were scaled down to 128×128 for the custom CNN. The transfer learning models, including VGG16 and ResNet-50, were pretrained on the ImageNet image classification data, which made it necessary to rescale the images to 224×224 pixels. For the custom CNN, a simple division normalization was applied to bring the pixel values into the 0-1 range. On the contrary, the channel-wise mean normalization was applied to the images for VGG16 and ResNet-50, using the pretrained weights. The data augmentation was performed only for the training set, as it is a standard practice for most machine learning tasks. The data was augmented randomly by rotating the images at a 15° angle, making width and height shifts, and applying horizontal flips and zooms. On the other hand, only resizing and normalization procedures were applied to the validation and test sets, as any other manipulations would make them biased.

2) Skin Disease Image Preprocessing: The dermoscopic images were resized as 224×224 pixels to make them suitable for usage by the EfficientNet and MobileNetV2 models in the skin disease classification module. For normalization, a mean of (0.485, 0.456, 0.406) and a standard deviation of (0.229,

0.224, 0.225) were used. The values for the calculation of the normalization parameters were obtained from the ImageNet. A comprehensive augmentation approach was adopted in the training stage in order to imitate the variability of the real-life conditions for clinical imaging. The implementation of different augmentations took into consideration changes related to the illumination, contrast and skin tone, random resizing from 0.85 to 1.0, random horizontal flipping, and color modification. Center-cropped resizing and normalization were applied only at the validation and testing stages in order to keep the correctness of the evaluation. The data were divided into training, validation, and testing in the ratio of 70:15:15 with the implementation of stratified sampling for letting all groups have the same distribution of classes.

C. Model Development

Once preprocessed images are available, the next challenge lies in picking and designing the architectures that will acquire the ability to recognize what the human eye is often blind to. Different model development and training strategies have been used for each disease module, which were each specific to their imaging domain.

1) Brain Tumor MRI Classification: Three distinct types of deep learning architectures were created and tested regarding brain MRI images classification across four types of tumors. The custom-built Convolutional Neural Network is composed of a two-block convolution in which the first one had 32 filters, and the second one had 64 filters, followed by batch normalization and max pooling, thus ending with a model featuring a fully connected classification part where dropout regularization technique was applied, and softmax was used for output of four classes. In addition two transfer learning methods – VGG16 and ResNet-50, which have been extensively analyzed and trained at ImageNet as well ended up being employed for achieving better classification results of brain MRI images. Both models were subjected to two-stage training process, wherein the first phase included freezing the convolutional base while training the classification head, and in the second phase, the backbone was unfrozen and fine-tuned at a lower learning rate for adapting it to MRI tumefaction-specific patterns.

The Adam optimizer was used in all examined models with a rate of 1×10^{-3}, a batch size of 32, and the categorical cross-entropy as a loss function. The custom CNN model was trained for 30 epochs and the validation loss served as a factor for the ReduceLROnPlateau process which reduces the learning rate by a factor of 2 if there is no change in the loss for three epochs. During Phase 1, the VGG16 model was trained for 15 epochs, and during Phase 2 it was retrained without EarlyStopping applied and with a learning rate of 1×10^{-5} over 15 epochs to avoid overfitting. ResNet-50 used a two-phase training process in which both phases involved the use of ReduceLROnPlateau to ensure that the model stays within a stable region of performance.

2) Skin Disease Classification: Three architectures were tested: EfficientNetB0, EfficientNetB3 and MobileNetV2, all pre-trained on ImageNet and fine-tuned for the five-class dermoscopic classification task. The ability of the compound scaling strategy in EfficientNet to take into account network depth, width and resolution simultaneously, makes it especially suitable for picking out the nuances of skin texture and morphology between skin conditions. MobileNetV2 has been added, which is based on depthwise separable convolutions, as a lightweight option for resource limited deployment environments. Pretrained backbone for all three architectures, then customized classification head (fully connected layer of 256 units, ReLU activation, Dropout layer (probability 0.5) and a five-class output layer) was added and trained end-to-end.

Each model was trained for 20 epochs, using the Adam optimizer and the initial learning rate of 1×10^{-3} and the batch size of 32, as the loss function, the cross-entropy. A StepLR scheduler was used to gradually decrease the learning rate by a factor of 0.1 every 7 epochs, which allowed the models to make big exploratory updates at the start of training and fine tuning updates closer to convergence, resulting in all five classes of skin disease benefiting from the increased stability and generalization performance.

D. Evaluation Metrics

A set of evaluation metrics was used to ensure a comprehensive and consistent evaluation of model performance for both the disease modules. Accuracy refers to the overall accuracy of all the results, whereas precision reflects the reliability of the positive prediction made by the model and Recall shows how well the model performs at identifying all the positive cases — an important measure when the implications of a misdiagnosis are serious like in a clinical setting. F1-Score is a balanced measure, which is the harmonic mean of Precision and Recall, particularly useful in cases of class imbalance. AUC-ROC is a measure of how well the model discriminates within the entire range of classification thresholds, and the one vs rest approach is used for multi-class evaluations. Confusion matrix was used to get a detailed classification-wise breakdown of the classification results. Further, to increase the transparency and trustworthiness of the proposed framework, Gradient-weighted Class Activation Mapping (Grad-CAM) was used as a visual interpretability tool, which produced saliency heatmaps for the image regions that play the most significant role in making the prediction in each of the models.

IV. Results and Discussion

A. Brain Tumor MRI Results

The performance of the proposed classification framework for brain tumors was assessed using a brain tumor dataset consisting of 7200 T1-weighted MRI scans across four clinically distinct classes: Glioma, Meningioma, No Tumor, and Pituitary Tumor, with three different deep learning architectures: custom Convolutional Neural Network (CNN), ResNet-50, and VGG16. The models were evaluated with the metrics namely accuracy, precision, recall, F1-Score, and confusion matrix analysis to get a multi-dimensional and comprehensive view of the classification performance.

The results are telling an obvious and continuous story of architectural evolution. The custom CNN, which was trained from scratch, provided a good baseline with 91% test accuracy. On a deeper look, however, its results were relatively poor in the class with a highest number of classes, Glioma, which is the most morphologically complex class with irregular boundaries and heterogeneous tissue composition that is a big hurdle for networks with shallow representational depth. The more complex the architecture, the higher the performance. ResNet-50, thanks to its residual connections which allow the network to learn deeper and more discriminative feature representations, achieved a test accuracy of 93% and a lesser misclassification in all four tumor categories. However, the breakthrough came with VGG16, which performed the best with a test accuracy of 94%, gaining the benefit of the deep hierarchical features learned by large-scale ImageNet pretraining and fine-tuned with the two-stage approach. Table I summarizes the comparative performance of all three models.

TABLE I
PERFORMANCE COMPARISON OF BRAIN TUMOR MRI CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score
CNN	91%	0.91	0.91	0.91
ResNet-50	93%	0.93	0.93	0.93
VGG16	94%	0.94	0.94	0.94

The detailed class-wise classification report for VGG16 is provided in Table II for further analysis of the strengths and weaknesses of the best performing model. Pituitary Tumors demonstrated a high Recall (1.00) and F1-Score (0.98) because pituitary tumors are highly distinct and localized in MR images. The No Tumor class also showed high performance, with a Recall of 0.99, indicating the model's ability to accurately predict healthy brain scans. The Glioma class achieved a high precision (0.99) but a low recall (0.82), suggesting that some glioma cases were incorrectly classified, as is known to be the difficulty of delineating a glioma from normal tissue in the same brain in a radiological sense. The overall performance of the Meningioma class was found to be balanced with Precision of 0.89, Recall of 0.95 and an F1-Score of 0.92.

TABLE II
VGG16 CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score
Glioma	0.99	0.82	0.89
Meningioma	0.89	0.95	0.92
No Tumor	0.93	0.99	0.96
Pituitary	0.95	1.00	0.98

The confusion matrix for the VGG16 model is shown in Fig. 1 which gives a detailed view of the model's classification results for each of the four categories. The data shows the near perfect diagonal concentration for the Pituitary Tumor and No Tumor classes, and confirms that the vast majority of MRI scans were correctly classified. The small misclassification rate between Glioma and Meningioma is clinically meaningful: Both tumor types may have similar intensity patterns in MRI and be difficult to differentiate due to irregular boundaries and edges, which is a well-known challenge in clinical radiology.

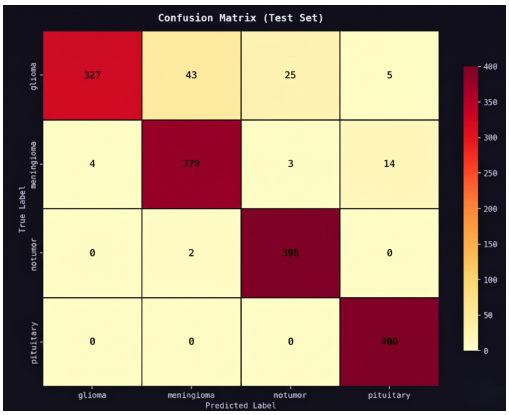


Fig. 1. Confusion Matrix of the VGG16 Brain Tumor MRI Classifier

In general, the outcomes of the experiments indicate that the transfer learning architectures considerably outperform models trained from scratch for the classification of brain tumor MRI images. VGG16 showed the best generalization capability for all four tumor types, confirming its potential to be used in the proposed AI-assisted diagnostic system for early brain tumor diagnosis and clinical decision support.

B. Skin Disease Results

The skin disease classification module was tested on a set of 9,548 dermoscopic images across five clinically different categories: Acne, Hyperpigmentation, Nail Psoriasis, SJS-TEN, and Vitiligo, using three different Deep Learning architectures: EfficientNetB0, EfficientNetB3, and MobileNetV2. The evaluation of each model was done using Accuracy, Precision, Recall, F1-Score, AUC-ROC and confusion matrix analysis to give a comprehensive, multi-dimensional evaluation of the classification performance.

The results show that the three architectures clearly have a performance hierarchy. EfficientNetB0 was the most successful, with a classification accuracy of 99% and an AUC of 0.993, a near perfect score which demonstrates the model's excellent performance of identifying subtle differences in both textural and morphological aspects between the different categories of skin diseases. The scaled-up version of the EfficientNet family, EfficientNetB3, achieved competitive results in accuracy (97.06%), proving that compound scaled architectures are very effective for dermoscopic image classification. In contrast, MobileNetV2 has a lower accuracy of 91%, but has proven itself in scenarios where resources are limited. The ROC curve comparison for all three models is shown in Fig. 2 and shows how well each architecture discriminates between the classes for different classification thresholds.

Table III shows the comparative results of all three models on all the evaluation metrics. EfficientNetB0 consistently outperformed all models in terms of Accuracy, Precision, Recall,

and F1-Score, thus showing better balance between sensitivity and reliability in the prediction of disease.

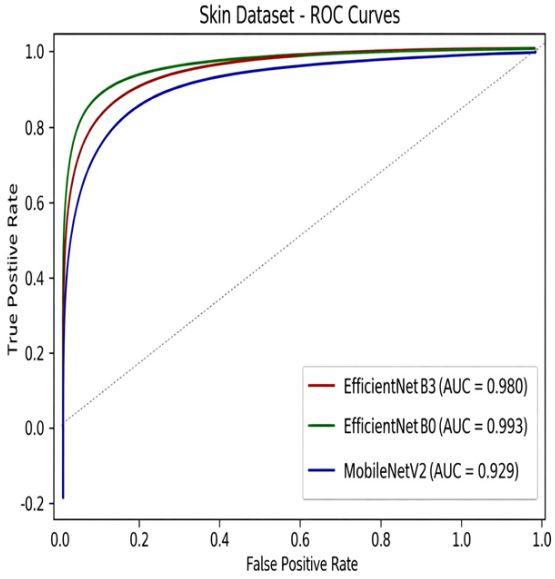


Fig. 2. ROC curve comparison of EfficientNetB0, EfficientNetB3, and MobileNetV2 for skin disease classification.

TABLE III
PERFORMANCE COMPARISON OF SKIN DISEASE CLASSIFICATION MODELS

Model	Accuracy	Precision	Recall	F1-Score
EfficientNetB0	0.99	0.99	0.99	0.99
EfficientNetB3	0.9706	0.9710	0.9706	0.9705
MobileNetV2	0.91	0.91	0.91	0.94

The class-wise performance of the best performing model is provided in Table IV to further investigate the classification behavior of the best performing model.To further investigate the classification behaviour of best performing model, the class-wise results of EfficientNetB0 is provided in Table IV. The model obtained almost 100% classification in all five disease categories. Nail Psoriasis had the highest Precision and Recall scores, both at 1.00, given the distinct and well defined lesion patterns which are evident in this condition. Acne also had a perfect Recall of 1.00 and F1-Score of 1.00. In spite of its strong scores in Precision (0.98) and Recall (0.98), hyperpigmentation had relatively lower scores than the other classes due to the following reasons: First, hyperpigmentation has a similar visual appearance to some hyperpigmented presentations of other skin diseases especially in terms of colour distribution and texture, meaning that the inter-class discrimination was more difficult. Second, hyperpigmentation was the smallest class, comprising only 109 images.

TABLE IV
CLASS-WISE PERFORMANCE OF EFFICIENTNETB0

Class	Precision	Recall	F1-Score	Support
Acne	0.99	1.00	1.00	173
Hyperpigmentation	0.98	0.98	0.98	105
Nail Psoriasis	1.00	1.00	1.00	225
SJS-TEN	1.00	0.99	0.99	225
Vitiligo	0.99	0.99	0.99	225
Overall Accuracy	0.99			

Fig. 3 shows the confusion matrix of EfficientNetB0, giving a granular view of the results of the classification across all five categories. As shown in the matrix, most samples were classified correctly, with a very few misclassified samples. In this study, there was a minor degree of confusion between the SJS-TEN and Nail Psoriasis categories, the latter having a similar trend to the former in terms of the texture and distribution of lesions when examining the dermoscopic images; clinically this is easily interpretable because both lesions share a common morphology.

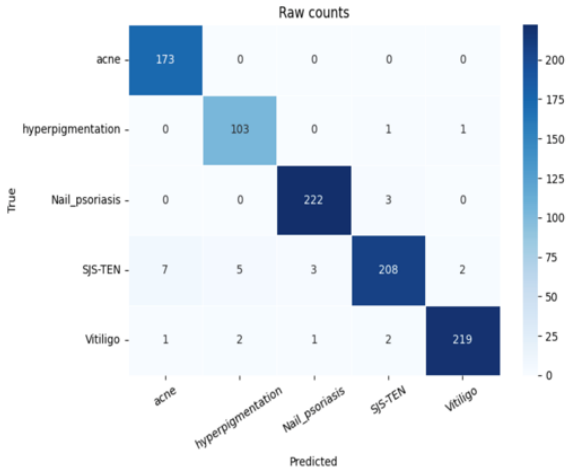


Fig. 3. Confusion matrix of EfficientNetB0 for skin disease classification.

For better model transparency and clinical interpretability, visualizations called "Grad-CAM" were created for EfficientNetB0. The activation heatmaps, as shown in Fig. 4 also showed that the model always predicted the skin areas of the infected region, not the background or non-informative parts of the images. The spatial coherence of the focus of the proposed model and affected areas of the disease bolsters its clinical validity and feasibility to be integrated into clinical workflows.

In summary, our results confirm EfficientNetB0 as the best architecture for classifying skin diseases in multi-class settings with high accuracy in all the metrics used. This is further bolstered by the integration of Grad-CAM, which adds a visual account of model predictions and makes the model a reliable and interpretable tool for the dermatological diagnosis with the assistance of AI.

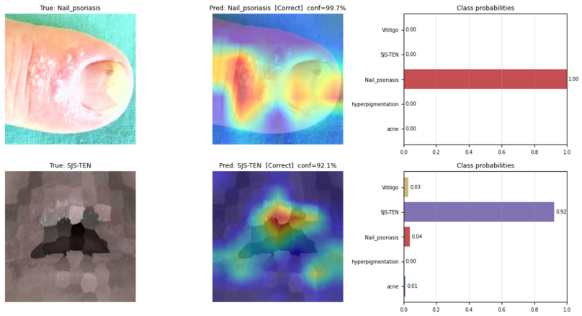


Fig. 4. Grad-CAM visualization for skin disease prediction using EfficientNetB0.

V. Conclusion and Future Work

According to the proposed method, the deep learning system that is being presented allows us to automate the process of categorizing brain tumors from MRI examinations and skin diseases from skin images. The best performance was achieved in brain tumors' categorization with a VGG16 model that produced a success rate of 94%. The second place was occupied by a standard CNN model whose accuracy was 91% and the third place went to ResNet-50 with a success rate of 93%. As for the skin disease categorization tasks, the work showed that the best results can be achieved with the help of EfficientNetB0 model with its accuracy of 99% and the AUC of 0.993. The use of Grad-CAM helped to improve the interpretability of predictions by providing visual explanations which highlight important areas of the brain and skin.

Future research will involve improvement of the proposed framework for clinical use and deployment. Future directions involve implementing federated learning for privacy-preserving distributed training, deploying to the cloud for real-time diagnostic support and inference to the edge for resource-limited clinical settings. Advanced explainability techniques, other than Grad-CAM [16], will be explored in the future, as well as fairness-aware machine learning techniques to mitigate demographic bias in diverse patient populations, and the integration of multimodal large language models for automated clinical report generation [17] to further enhance the practical utility and clinical reliability of the proposed framework.

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